# TENDAI N.R. GWANZURA, PhD (c)

*PhD Candidate, Epidemiology | NIH F31 Fellow | Health Disparities | Infectious Disease | Population Health Analytics*
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## PROFESSIONAL SUMMARY

PhD candidate in Epidemiology at Florida International University (expected Winter 2026) and NIH Ruth L. Kirschstein NRSA F31 Fellow with over ten years of combined scientific experience in epidemiological research, population health analytics, and public health program evaluation. My work spans HIV and COVID-19 outcomes research, health disparities, behavioral epidemiology, spatial analysis, and infectious disease surveillance. I have contributed to federally funded research programs producing peer-reviewed publications, led data collection and analysis across large administrative and behavioral health datasets, and applied a range of statistical methods to public health questions. I am proficient in SAS, R, Python, SQL, and ArcGIS Pro, and I bring experience in both academic research and applied public health settings.

## CORE SKILLS

**Epidemiological Methods:** Cohort study design, causal inference (DiD, IPW, propensity score methods), survival analysis (Cox, competing risks, Firth logistic regression for rare events), multilevel modeling, longitudinal and panel data analysis, spatial epidemiology (Getis-Ord Gi*, LISA, hotspot analysis), complex survey analysis, behavioral epidemiology, EMA methodology

**Statistical Software:** SAS (advanced), R (tidyverse, survival, lme4, lavaan, ggplot2, gtsummary, Quarto), Python (pandas, NumPy, statsmodels, scikit-learn, TensorFlow), Stata, SQL, ArcGIS Pro, Tableau, SPSS

**Research Skills:** IRB protocol development, study design, data collection management, primary and secondary data analysis, analytic plan development, federal progress reports, peer-reviewed manuscript authorship, scientific communication

**Data Sources:** EHR-linked administrative health registries, NHANES, NSDUH, BRFSS, state surveillance registries, behavioral survey instruments, REDCap

## PROFESSIONAL EXPERIENCE

**Graduate Assistant / Doctoral Researcher** | *Florida International University, Department of Epidemiology* May 2025 – Present

- Teaching graduate-level epidemiological methods to MPH and PhD students, developing course materials and providing individual statistical consultation to students at varying levels of quantitative preparation.
- Completing doctoral dissertation comprising three aims: causal analysis of COVID-19 mortality disparities before and after vaccine availability among people with HIV; spatial analysis of HIV and COVID-19 hotspot co-occurrence; and longitudinal policy analysis of county-level COVID-19 prevention strategies and incidence trends.
- Advancing three first-author manuscripts through peer review simultaneously while completing dissertation requirements.
- Mentored one junior researcher on epidemiological methods and graduate school application preparation; that mentee successfully applied to and was accepted into a graduate program.

**National Teaching Fellow in Data Science** | *RCMI National Coordinating Center, Morehouse School of Medicine* Nov 2024 – Present

- Designed and delivered graduate-level data science and statistical programming instruction in R through the RCMI National Coordinating Center, serving MPH, PhD, and early, mid, and late-career researchers across multiple minority-serving institutions including Morehouse School of Medicine and Howard University.
- Curriculum covered reproducible research, regression modeling, data visualization, and statistical communication using Quarto and R.

**NIH F31 Predoctoral Fellow** | *Florida International University & National Institutes of Health (NRSA Award, $76,000)* May 2023 – May 2025

- Led an independent federally funded research program evaluating COVID-19 outcomes and mortality disparities among people with HIV in Florida, applying difference-in-differences with inverse probability weighting and propensity score methods to a 2M+ record EHR-linked statewide administrative registry.
- Applied Firth logistic regression for rare-event outcomes in a dissertation manuscript examining individual and community-level determinants of HIV and COVID-19 hotspot co-occurrence across Florida counties.
- Conducted longitudinal policy analyses examining associations between county-level COVID-19 prevention strategies and COVID-19 incidence among people with and without HIV, using county-by-day panel data structures with policy indicator variables.
- Built and maintained reproducible analytical pipelines in R and SAS using Git version control; constructed multi-table longitudinal cohorts from administrative health data with systematic data quality validation.
- Produced all federal deliverables including analytic plans, progress reports, and technical summaries on schedule throughout the two-year fellowship.

- Presented findings at the Society for Epidemiologic Research Annual Meeting 2024 (oral) and APHA 2023 (oral); received the Susser-Stein Award for Methodological Excellence, SER 2024.

**Graduate Research Analyst** | *Florida International University, Dept. of Epidemiology (NIH R01 & K-award)* Aug 2021 – Aug 2023

- Contributed statistical analyses to a $2M+ NIH-funded research portfolio on HIV treatment effectiveness, ART adherence, and healthcare engagement in racially diverse populations in Miami-Dade County, generating three peer-reviewed publications.
- Co-designed, coded, and analyzed the Project D.A.I.L.Y. Phase 2 baseline questionnaire, incorporating ten validated behavioral and psychosocial instruments including the TSRQ, HIV-ASES, Brief COPE, CES-D, GAD-7, and HIV Stigma scales, and applied Ecological Momentary Assessment methodology to characterize daily behavioral factors associated with ART adherence.
- Applied Causal Bayesian Networks and structural equation modeling to assess patient-provider relationship effects on HIV viral suppression, co-authoring a peer-reviewed publication in AIDS and Behavior.
- Co-led primary data collection from approximately 1,000 community participants across Miami-Dade County, coordinating across investigators and community partners while maintaining IRB compliance and data quality standards throughout.
- Applied LASSO-regularized regression to NHANES complex survey data to identify predictors of health outcomes; built reproducible SAS and R programs with complete documentation.

**Quantitative Modeling Fellow, Data Science for Women** | *Correlation One (2–5% acceptance)* Jun 2022 – Aug 2022

- Selected through a competitive 2 to 5% acceptance process to analyze NHANES federal health survey data (approximately 60,000 individuals, 1999 to 2020) as part of a six-person team examining variables associated with organophosphate pesticide exposure.
- Applied LASSO regression to combined demographic, dietary, and laboratory data to identify the most important features associated with organophosphate levels; urine creatinine and blood lipid values emerged as key predictors across multiple organophosphate outcomes.
- Contributed data cleaning, feature engineering, exploratory data analysis, and visualization; presented findings and policy recommendations to program cohort and leadership.

**DS4A Fellow, Data Science for All** | *Correlation One (Team 47)* Nov 2021 – Apr 2022

- Collaborated on a three-person team to develop a convolutional neural network using TensorFlow and Keras to classify brain MRI images as tumor or no tumor, trained on 6,123 standardized images sourced from Kaggle, achieving 97.79% accuracy and validation loss of 0.15 after 10 training epochs.
- Built and deployed a live interactive dashboard using the Anvil platform, enabling remote MRI image upload with real-time tumor probability output, providing a tool to support faster and more accessible brain tumor detection.

**Health Program Analyst** | *LA County Department of Public Health* Nov 2020 – Jul 2021

- Analyzed electronic health record data evaluating vaccination coverage and health program outcomes for people experiencing homelessness and other medically vulnerable populations across LA County facilities, building SAS-based dashboards tracking population health metrics for program leadership.
- Produced over 200 line lists in a single week during peak pandemic response operations, supporting real-time outbreak monitoring and resource allocation decisions for county program leadership.
- Validated and integrated healthcare data from multiple administrative sources, maintaining data accuracy for surveillance reporting and communicating findings to clinical and program staff.

**Data Analyst Intern — Stop Diabetes Program** | *Community Translational Research Institute (CITRI), Pomona, CA* Feb 2020 – Aug 2020

- Monitored and supported data collection processes for a community-based Diabetes Prevention Program serving underserved populations in Pomona, California.
- Cleaned and validated electronic tracking system records, performed statistical analyses using SAS and SPSS, and prepared weekly data analysis reports for program leadership.
- Contributed to program monitoring activities and assisted with data management and quality control throughout the internship period.

**Medical Laboratory Scientist** | *CIMAS Medical Labs, Harare, Zimbabwe* Dec 2014 – Apr 2018

- Managed clinical patient records and entered laboratory results across hematology, immunology, microbiology, and biochemistry in a high-volume HIV clinical setting, developing hands-on experience with clinical healthcare data systems and patient record integrity.
- Led ISO accreditation audits and developed standard operating procedures and quality control documentation, training multidisciplinary staff on quality systems across all laboratory workflows.
- Processed clinical samples and interpreted results daily in a high-burden infectious disease environment, developing foundational scientific training in immunology and hematology.

## EDUCATION

**PhD in Public Health, Epidemiology** | Florida International University, Miami, FLExpected Winter 2026 *Dissertation: COVID-19 outcomes, health disparities, spatial epidemiology, and policy analysis in people with HIV*

**Master of Public Health (MPH)** | Claremont Graduate University, Claremont, CA2020 *Health behavior theory, biostatistics, social determinants of health*

**BSc (Hons) Medical and Laboratory Science** | University of Zimbabwe2014 *Immunology, hematology, microbiology, biochemistry | MLS Certification*

## PUBLICATIONS — 13 PEER-REVIEWED | 3 FIRST-AUTHOR UNDER REVIEW

- Sheehan DM, Gwanzura TNR, et al. Daily Factors Associated with ART Adherence Among Young Latino Sexual Minority Men with HIV. AIDS and Behavior. 29:3054-3069 (2025).
- Trepka MJ, Gwanzura TNR, et al. Using Causal Bayesian Networks to Assess Patient-Provider Relationship Effects on HIV Viral Suppression. AIDS and Behavior. 28(6):2113-2130 (2024).
- Ward MK, Gwanzura TNR, et al. Nonfatal Opioid-Related Overdoses Before and During COVID-19 in Florida. Preventive Medicine Reports. 31:102102 (2023).
- Gwanzura TNR, et al. The Effect of Vaccine Availability on Disparities in COVID-19 Mortality Among People with HIV. Revise & Resubmit, PLoS ONE (2026).
- Gwanzura TNR, et al. Hotspot Analysis of HIV and COVID-19 Co-Occurrence in Florida. Under Review, Health and Place (2026).
- Full list available at ORCID: 0000-0002-2110-3530 | Google Scholar: scholar.google.com/scholar?q=tendai+gwanzura

## AWARDS & CERTIFICATIONS

NIH Ruth L. Kirschstein NRSA F31 Fellowship ($76,000, 2023-2025) | Susser-Stein Award for Methodological Excellence, Society for Epidemiologic Research (2024) | Delta Omega Honorary Society in Public Health (2020) | CITI Good Clinical Practice (GCP) Certification

## PROFESSIONAL LEADERSHIP & SERVICE

**Governing Councilor** | American Public Health Association (APHA) 2025 – Present
**President, Students and Post-docs Committee (SPC)** | Society for Epidemiologic Research 2026 – Present
**National Chapter Chair** | Sisters in Public Health (300+ members) 2024 – 2026
**Abstract Reviewer** | Society for Epidemiologic Research Annual Meeting